

## EDUCATION

### McGill University, Schulich School of Music

Ph.D. in Music Technology

Sep. 2022 - Present

Montreal, Canada

**Supervisors:** Prof. Marcelo M. Wanderley and Assoc. Prof. Isabelle Cossette

### Shanghai Jiao Tong University, USC-SJTU Institute of Cultural and Creative Industry

M.A. in Cultural & Creative Industry Management

Sep. 2019 - Mar. 2022

Shanghai, China

**Advisors:** Asst. Prof. Gus Xia and Asst. Prof. Zhixin Xu

### Peking University, National School of Development

Bachelor in Economics, Double Degree

Sep. 2017 - Jun. 2019

Beijing, China

### Beijing University of Posts and Telecommunications, School of Digital Media

B.E. in Digital Media Technology

Sep. 2015 - Jun. 2019

Beijing, China

## RESEARCH INTERESTS

My research interests center around music and human-computer interaction, with a particular focus on musical haptics and embodied interaction. By researching music, I aim to inspire the broader HCI field with new paradigms of multimodal interaction, demonstrating how sensing, haptics, and AI-driven modeling of embodied musical practices can inform the design of more intuitive, expressive, and socially connected interactive systems.

## PEER-REVIEWED PUBLICATIONS

### Journal Articles

- [J.1] **Ziyue Piao**, Christian Frisson, Bavo Van Kerrebroeck, and Marcelo M. Wanderley, "**Assessing the Impact of Force Feedback in Musical Knobs on Performance and User Experience**", *Actuators*, 2024.

### Conference Full Papers

- [C.3] **Ziyue Piao**, Yohei Wada, Isabelle Cossette, Marcelo M. Wanderley, and Akira Maezawa, "**Exploring Breathing-Music Coupling: Using the Breathing Mirror for Somatic Reflection in Piano Performance**", *International Conference on New Interfaces for Musical Expression (NIME)*, London, UK, 2026.
- [C.2] Leon Lu, Chase Crispin, **Ziyue Piao**, Aino Eze-Anyanwu, and Audrey Girouard, "**Project TapTap: A Longitudinal Study Exploring Non-Verbal Communication through Vibration Signals Between Teachers and Blind or Low Vision Music Learners**", *ACM International Conference on Human Factors in Computing Systems (CHI)*, Yokohama, Japan, 2025
- [C.1] **Ziyue Piao**, Marcelo M. Wanderley, and Felipe Verdugo, "**MappEMG: Enhancing Music Pedagogy by Mapping Electromyography to Multimodal Feedback**", *International Conference on ArtsIT, Interactivity and Game Creation (ArtsIT)*, Campinas, Brazil, 2023.

### Conference Short Papers

Note: All types of publications except full papers

- [P.5] **Ziyue Piao**, Isabelle Cossette, and Marcelo M. Wanderley, "**Reconciling Kinesthetic Mismatches: A Somatic Alignment Mindset for Musical Body Transformation**", *ACM International Conference on Human Factors in Computing Systems (CHI)*, Barcelona, Spain, 2026.
- [P.4] **Ziyue Piao** and Kanyu Chen, "**Respiratory Musical Interfaces (RMIs) in NIME**", *International Conference on New Interfaces for Musical Expression (NIME)*, London, UK, 2026.
- [P.3] **Ziyue Piao**, "**Exploring Haptics to Augment Kinesthetic Learning in Singing**", *ACM International Conference on Tangible, Embedded, and Embodied Interaction (TEI)*, Bordeaux, France, 2025.
- [P.2] **Ziyue Piao** and Gus Xia, "**Sensing the Breath: A Multimodal Singing Tutoring Interface with Breath Guidance**", *International Conference on New Interfaces for Musical Expression (NIME)*, Online, 2022.
- [P.1] Yinmiao Li, **Ziyue Piao**, and Gus Xia, "**A Wearable Haptic Interface for Breath Guidance in Vocal Training**", *International Conference on New Interfaces for Musical Expression (NIME)*, Shanghai, China, 2021.

### Yamaha Corporation

Research Collaboration, Mentors: Dr. Akira Maezawa and Yohei Wada

Sep. 2025 - Present  
Hamamatsu, Japan and Remote

#### Breathing Analysis with Sensing Wearables for Piano Performance:

- Led an in-lab study with seven pianists, capturing synchronized piano performance and respiration data through Yamaha GummiStra wearable breathing belts.
- Built an annotated visual interface that layers multimodal breathing and performance signals to streamline note taking and playback.
- Synthesized qualitative interview findings with quantitative breathing and performance metrics to surface design insights.
- Prototyped and iteratively refined the corset with GummiStra sensor to measure thoracic expansion during music performances.

### Sony Computer Science Laboratories

Research Intern, Mentors: Dr. Shinichi Furuya, Asst. Prof. Jun Nishida (University of Maryland)

Jun. 2025 - Sep. 2025  
Tokyo, Japan

#### ScapuLearn: Finger Dynamics Estimation through the Scapular Muscle-skeletal Movements:

- Developed a machine-learning pipeline that process shoulder-mounted IMU data to estimate finger force, loudness, keyboard register, and scapular posture during piano pieces.
- Analyzed shoulder's rotation and acceleration features and trained gradient-boosted models to infer finger-level dynamics from minimally intrusive upper-body sensing.
- Validated the approach through multi-stage user studies with twenty pianists, demonstrating how wearable sensing plus ML can support performance analysis and pedagogy.

### Input Devices and Music Interaction Lab & Music Performance and Body Lab, McGill University

Research Assistant, Advisors: Prof. Marcelo M. Wanderley and Assoc. Prof. Isabelle Cossette

Sep. 2022 - Present  
Montreal, Canada

#### Challenges and Design Opportunities in Teaching Singing Breath Control:

- Led focus groups with eleven teachers and students to map kinesthetic pain points in teaching breath control and prioritize assistive technology concepts.

#### Haptic Music Editing Interface for Pitch Modulation:

- Built a web interface that streams TorqueTuner haptic data into a pitch-modulation synthesizer for real-time editing.
- Conducted controlled studies on how various force-feedback modes affect fine-grained vibrato control.
- Analyzed both pitch-contour performance data and qualitative feedback via Likert-scale ratings and interviews.

### Creative Interactions Lab, Carleton University

Visiting Researcher, Advisors: Prof. Audrey Girouard, Leon Lu

May. 2024 - Aug. 2024  
Ottawa, Canada

#### Wearable Haptics for Music Accessibility:

- Performed thematic analysis of a ten-week study with twelve music teachers and visually impaired students to document how a wearable vibrotactile tool expands music-accessibility strategies.

### Laboratory of Simulation and Movement Modelling, University of Montreal

Research Intern, Advisors: Prof. Felipe Verdugo and Prof. Marcelo M. Wanderley

Sep. 2022 - Aug. 2023  
Montreal, Canada

#### MappEMG: Transmitting Muscle Effort to Vibration for Piano Pedagogy:

- Implemented real-time muscle-effort capture by integrating Delsys and BITalino EMG streams into a single calibration pipeline.
- Developed the MappEMG system to map EMG data to smartphone vibration for teacher-student feedback.
- Led evaluation workshops in five music conservatories to assess its pedagogical impact with more than forty piano learners.

### Computer Music Group, Carnegie Mellon University

Research Assistant, Advisor: Prof. Roger B. Dannenberg

Mar. 2021 - Mar. 2022  
Remote

#### Music Computing and Expectation Modeling:

- Investigated predictive models for musical sequences in contemporary music.
- Enhanced expectation-network algorithms by replacing simple counting with gradient-descent-based learning.

### Music X Lab, New York University Shanghai

Research Assistant, Advisor: Asst. Prof. Gus Xia

Mar. 2020 - Jan. 2022  
Shanghai, China

#### Multimodal Vocal Tutoring System:

- Developed an interactive score that plots breathing traces alongside pitch notation so singers can align phrasing with airflow.
- Integrated airflow, chest-expansion, and audio sensors to detect breathing patterns in real time during lessons.
- Prototyped a wearable device to track dynamic breathing movements.

## ACHIEVEMENT

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### Funding and Grants

- **Student Award** (Collaborator: Hanwen Zhang), CIRMMT, \$8,000 CAD. 2025
- **TCH Innovation in Haptics Research Grant**, IEEE Technical Committee on Haptics, \$ 2,300 USD. 2024
- **FRQSC Doctoral Scholarship**, Fonds de recherche du Québec, \$ 83,334 CAD (Top 1/21). 2024 - 2028
- **Student Engagement Fund Winner** (Collaborators: Alberto Acquilino, Theodora Nestorova), McGill University, \$1,500 CAD. 2024
- **Inter-centre Research Exchange Funding**, CIRMMT, \$3,000 CAD. 2023
- **Student Award** (Collaborator: Theodora Nestorova), CIRMMT, \$10,000 CAD. 2023

### Research Awards

- **ISMIR WIMIR Accomodation & Registration Grants**, International Society for Music Information Retrieval Conference. 2025
- **Best Oral Presentation**, The 2nd International Electronic Conference on Actuator Technology (IECAT). 2024
- **Best Oral Presentation**, COBS Student Colloquium. 2024
- **Graduate Excellence Award**, McGill University. 2022 - 2025
- **Online Conference Award**, CIRMMT. 2023 - 2024
- **Graduate Research Enhancement and Travel (GREAT) Award**, McGill University. 2023
- **NIME Ableton Scholarship**, International Conference on New Interfaces for Musical Expression. 2022
- **ISMIR Registration Grant**, International Society for Music Information Retrieval Conference. 2021

### Other Honors

- **Differential Fee Waiver for International Students**, McGill University. 2022 - 2023
- **Outstanding Graduate**, Shanghai Jiao Tong University (Top 1%). 2022
- **Outstanding Teaching Assistant Award**, Shanghai Jiao Tong University (Top 1%). 2020
- **Excellent Teaching Assistant Award**, Foundational General Education of Shanghai Jiao Tong University. 2020
- **Outstanding Graduate of Beijing**, Beijing University of Posts and Telecommunications (Top 3%). 2019
- **Honorable Prize**, International Mathematical Contest in Modeling. 2018
- **Hengtong Photoelectric Enterprise Scholarship**, Beijing University of Posts and Telecommunications. 2017
- **1st Prize Academic Excellence Award**, Beijing University of Posts and Telecommunications. 2016, 2018

## TEACHING

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### McGill University

*Co-instructor with Champ Darabundit*

- MUMT 302 New Media Production 1 2025 Winter
- MUMT 302 New Media Production 1 2024 Winter

*Guest Lecturer*

- MUPP 693 Performance Practice Seminar: *Feel the Music: Haptics in Music Learning* 2026 Winter
- MUPD 235 Music as a Profession: *Musical Haptics and Musicians* 2025 Winter
- MUMT 620 Gestural Control of Sound Synthesis: *Designing a Multimodal Interface for Music Pedagogy* 2023 Fall

### New York University

*Guest Lecturer*

- CSCI-SHU 188 Introduction to Computer Music: *How to Design a Singing Tutoring Interface* 2021 Fall
- CSCI-SHU 188 Introduction to Computer Music: *Design a Single Thread System — a Singing Tutoring UI Demo* 2020 Fall

### Shanghai Jiao Tong University

*Teaching Assistant*

- Introduction to Computer Music 2021 Winter
- Symphony 2020 Fall
- Chinese and Western Music Theory 2020 Spring

## MENTORING

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- **Jiayi Liang (2025), MS in Innovation Design Engineering, Imperial College London:** Vocal Flux: Visualizing your Voice through Muscle Waves
- **Gabrielle Seymour (2023-2024), BS in Computer Science, McGill University:** Audio-Visual Interactive for Ambient Music Generation

**Committee Leadership Roles**

|                                                                                                            |             |
|------------------------------------------------------------------------------------------------------------|-------------|
| ◦ NIME 2027 Design Chair                                                                                   | 2026 - 2027 |
| ◦ ISMIR 2026 Volunteer Chair                                                                               | 2025 - 2026 |
| ◦ IEEE Technical Committee on Haptics (TCH) Student Representative                                         | 2025 - 2026 |
| ◦ ACM International Conference on Tangible Embedded and Embodied Interaction (TEI) 2026 Registration Chair | 2025 - 2026 |
| ◦ ACM TEI 2026 Associate Chair of Work-In-Progress Track                                                   | 2025 - 2026 |
| ◦ CIRMMT Student Representative                                                                            | 2024 - 2026 |
| ◦ CIRMMT Student Coordinator for Research Axis 3                                                           | 2023 - 2024 |
| ◦ Shanghai Jiao Tong University (SJTU) Academic Affair Fellow                                              | 2021 - 2022 |

**Academic Reviews**

*Reviewed 47 submissions total. Years marked with \* indicate Special Recognitions for Outstanding Reviews.*

**Journals**

|                                                |      |
|------------------------------------------------|------|
| ◦ IEEE Transactions on Haptics (ToH), article. | 2026 |
| ◦ Springer Nature Scientific Reports, article. | 2026 |

**Conferences**

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|---------------------------------------------------------------|--------------|
| ◦ NIME, Papers and Workshops.                                 | 2021–2026    |
| ◦ ACM CHI, Papers and Late Breaking Works.                    | 2025*, 2026* |
| ◦ ACM Design Interactive Systems (DIS), Papers.               | 2025         |
| ◦ ACM TEI, Papers and Pictorials.                             | 2025, 2026   |
| ◦ ACM Creativity & Cognition (C&C), Papers.                   | 2025         |
| ◦ ACM Ubiquitous Computing (UbiComp-ISWC), Demos and Posters. | 2024, 2026   |
| ◦ EAI ArtsIT, Interactivity & Game Creation (ArtsIT), Papers. | 2024         |

**Volunteer**

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| ◦ Volunteer of IEEE Haptic Symposium 2026 | 2026 |
| ◦ Volunteer of ISMIR 2024                 | 2024 |
| ◦ Senior volunteer of NIME 2021           | 2021 |

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**TALKS AND PRESENTATIONS**

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|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| • Music Learning with Haptics: Internal Awareness, Motor Control, and Collaborative Metaphor, <i>CIRMMT Haptics Workshop</i> .          | Apr. 2026 |
| • The Duality of Breath: From Physiological Spontaneity to Conscious Control in Musical Expression, <i>McGill Doctoral Colloquium</i> . | Apr. 2026 |
| • How does breathing convey expressive and musical dynamics in piano performances, <i>Yamaha Corporation</i> .                          | Dec. 2025 |
| • Touching the Unseen: Enhancing Music Learning with Wearable Haptics, <i>Yamaha Corporation</i> .                                      | Sep. 2025 |
| • ScapuLearn: Minimally-intrusive Estimation of Finger and Postural Dynamics via Scapular Musculoskeletal Sensing, <i>Sony CSL</i> .    | Aug. 2025 |
| • Bridging Kinesthetic Mismatches in Music Learning with Wearable Haptic Devices, <i>COBS Student Colloquium</i> .                      | May. 2025 |
| • CantusPlay: An Interactive Singing Interface for Practicing Improvised Renaissance Counterpoint, <i>COBS Student Colloquium</i> .     | May. 2025 |
| • Exploring on Breathing Challenges and Technology Assisted Solutions, <i>The Voice Foundation Annual Symposium</i> .                   | May. 2025 |
| • How to write a success CIRMMT student award proposal, <i>CIRMMT Colloquial Colloquium</i> .                                           | Jan. 2025 |
| • Assessing the Impact of Force Feedback in Musical Knobs on Performance and User Experience, <i>IECAT Conference 2024</i> .            | Nov. 2024 |
| • SenSing: Transmitting Kinesthetic Breathing Experiences for Voice Pedagogy, <i>PAVA Symposium 2024</i> .                              | Oct. 2024 |
| • Preliminary Exploration of Kinesthetic Breathing Feedback for Enhanced Vocal Pedagogy, <i>COBS Student Colloquium</i> .               | May 2024  |
| • Designing a Multimodal Interface for Music Pedagogy, <i>CIRMMT Workshop with Prof. Karon MacLean</i> .                                | Nov. 2023 |
| • SenSing: Transmitting Kinesthetic Breathing Experiences for Voice Pedagogy, <i>CIRMMT MMR Re-Open Celebration</i> .                   | Oct. 2023 |
| • SenSing: Transmitting Kinesthetic Breathing Experiences for Voice Pedagogy, <i>COBS Student Colloquium</i> .                          | May 2023  |
| • MappEMG – Feeling the muscle activity of classical musicians, <i>CIRMMT Workshop on Piano Knowledge Transfer</i> .                    | May 2023  |
| • Sensing the Singing Body, <i>CIRMMT Workshop with Pamela Z</i> .                                                                      | Jan. 2023 |
| • Sensing the Breath: Multimodal Singing Tutoring Interfaces with Breath Guidance, <i>Music + AI Reading Group, Mila</i> .              | Oct. 2022 |
| • Multimodal Singing Tutoring Interfaces Design for Breathing Guidance, <i>IDMIL, McGill University</i> .                               | Sep. 2022 |

## ARTISTIC PERFORMANCES AND ROLES

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### Interactive Artworks & Residencies

- **Somatic Resonances for Soprano and Ambient Electronics:** *Composer and Augmented Vocal Performer* June 2027  
Bled Contemporary Music Week, Bled, Slovenia.
- **Dao:** *Laptop Orchestra (Composer, Director, & Performer)* Dec. 2020  
Sprout Art Exhibition, Shanghai, China.

### Classical Music Performances

- **C. (F. Poulenc):** *Soprano Soloist* May 2025  
Living Your Music Seasonal Recital, Montreal, Canada.
- **He Shall Feed His Flock** (G. Handel): *Soprano-Alto Duet* Dec. 2024  
Living Your Music Seasonal Recital, Montreal, Canada.
- **Se tu m'ami** (A. Parisotti): *Soprano Soloist* Dec. 2023  
Living Your Music Seasonal Recital, Montreal, Canada.
- **ECNU Alma Mater Song (Arranged):** *Vocal Chamber Quartet (Alto)* Oct. 2021  
Featured on the NYU Shanghai Official WeChat Channel, Shanghai, China.
- **Huo Ba Jie, Red Sorghum, & Kung Fu Panda:** *Orchestra Guzheng Player* Dec. 2019  
Jiang Xin Concert, SJTU Orchestra, Shanghai, China.

### Artistic Leadership and Roles

- **Vocal Artist (Soprano):** Advanced classical training with Dr. Irene M. Feher and Giancarlo Scalia. 2022 - Present
- **Chamber Vocal Artist (Alto):** Performed polyphonic choral works with SJTU Chamber Choir. 2020 - 2021
- **Principal Guzheng Instrumentalist:** Performed with SJTU Chinese-Western Orchestra. 2019 - 2020
- **Artistic Director:** Curated cross-disciplinary stage productions at School of Digital Media. 2015 - 2019

## SKILLS

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- **Programming:** Python\*, C++, Javascript\*.
- **Audio Production:** Logic Pro\*, Audition, PureData\*, MAX MSP
- **Prototyping:** Arduino\*, Fusion 360, 3D Printing.
- **VR, AR and Game:** Maya\*, Processing\*, Unity, After Effects.
- **Graphic Design:** Illustrator\*, Photoshop\*.
- **Toolkits:** scikit, PyTorch.
- **Languages:** Mandarin (native), English (advanced), French (beginner).

Items marked with \*: professional proficiency; others: used at least once in projects.

## REFERENCES

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|                                       |                            |                                |
|---------------------------------------|----------------------------|--------------------------------|
| <b>Prof. Marcelo M. Wanderley</b>     | McGill University          | marcelo.wanderley@mcgill.ca    |
| <b>Assoc. Prof. Isabelle Cossette</b> | McGill University          | isabelle.cossette1@mcgill.ca   |
| <b>Prof. Roger B. Dannenberg</b>      | Carnegie Mellon University | rbd@cs.cmu.edu                 |
| <b>Assoc. Prof. Gus Xia</b>           | MBZUAI                     | gus.xia@mbzuai.ac.ae           |
| <b>Dr. Akira Maezawa</b>              | Yamaha Corporation         | akira.maezawa@music.yamaha.com |
| <b>Dr. Irene M. Feher</b>             | Concordia University       | irene.feher@concordia.ca       |

Available upon request.